# Conjecture 00000001741: disproof of the stated upward optimality

C0ldSmi1e

October 12, 2026

**Abstract**

The conjecture asserts infinitude of positive integer pairs satisfying a cube–square gap bound, together with the assertion that its logarithmic exponent cannot be increased. We prove that infinitude for this upper bound is preserved by every increase of the exponent. Consequently the two stated clauses cannot both hold. This does not decide the infinitude clause by itself, Hall's conjecture, or a corrected optimality assertion involving a decrease of the exponent.

## 1 Exact assertion and formal interpretation

The English conjecture reads:

> The reverse direction of Hall's conjecture: there exist infinitely many pairs with $0 < |x^3 - y^2| < \sqrt{x}\,(\log x)^{1/4}$; the logarithmic correction is optimal (the exponent $1/4$ cannot be increased).

The Chinese version also explicitly says that the exponent cannot be increased. The exact bilingual source is included as `conjecture.md`. Both variables are positive integers, as specified in the source's definition paragraph. That paragraph provides background; no form of Hall's conjecture is assumed as a hypothesis in this proof.

We use the natural logarithm and Mathlib's real powers. At boundary values, these total functions use $\log 0 = 0$, $0^0 = 1$, and $0^a = 0$ for $a \neq 0$. This makes the following definitions meaningful for every natural coordinate and real exponent. The conjecture itself uses only positive coordinates and positive exponents, where the usual nonnegative-base interpretation applies. For $a \in \mathbb{R}$, write

$$B_a(x) = \sqrt{x}\,(\log x)^a, \qquad D(x,y) = |x^3 - y^2|,$$

$$S_a = \{(x,y) \in \mathbb{N}^2 : x > 0,\ y > 0,\ 0 < D(x,y) < B_a(x)\}, \qquad I(a) \iff S_a \text{ is infinite.}$$

Subtraction in $D$ is over $\mathbb{R}$, after casting the natural-number coordinates; it is not truncated subtraction. The condition $D > 0$ excludes exact cube–square coincidences. The statement is infinitude of distinct pairs.

The parenthetical specifies the direction of optimality. Its literal reading in the displayed infinitude assertion gives the conjunction

$$C = I(1/4) \ \wedge\ (\forall b \in \mathbb{R},\ b > 1/4 \implies \neg I(b)). \tag{1}$$

We prove $\neg C$. The witness exponent will be $b = 1$, so allowing only rational larger exponents would not change the disproof. We do not replace "increased" by "decreased" or move the logarithm into a denominator.

## 2 Finite exceptional pairs

Increasing the exponent need not increase the bound when $0 < \log x < 1$. Those small first coordinates must be treated separately, including their possible second coordinates.

**Lemma 1.** *For every real $a$ and fixed $x \in \mathbb{N}$, the set of pairs in $S_a$ with first coordinate $x$ is finite. Consequently, for every $N \in \mathbb{N}$, the set of pairs in $S_a$ with $x < N$ is finite.*

*Proof.* Choose a natural number $M > x^3 + B_a(x)$, using the Archimedean property of $\mathbb{R}$. If $(x, y) \in S_a$, the strict absolute-value bound implies

$$y^2 < x^3 + B_a(x) < M.$$

Since $y$ is a positive integer, $y \leq y^2$, hence $y < M$. Thus this fiber is contained in the finite set $\{(x, y) : y \in \mathbb{N},\ y < M\}$. There are finitely many $x < N$, so the union of these finite fibers is finite. □

**Lemma 2.** *If $I(a)$ holds, then for every natural $N$ the set $\{(x, y) \in S_a : N \leq x\}$ is infinite.*

*Proof.* Remove the finite set from Lemma 1 from the infinite set $S_a$. The remaining pairs have $x \geq N$. □

## 3 Increasing the exponent preserves infinitude

**Theorem 1.** *For all real $a \leq b$, $I(a)$ implies $I(b)$.*

*Proof.* Choose a natural $N \geq \exp(1)$. For $x \geq N$, monotonicity of the logarithm gives $\log x \geq 1$. Real exponentiation with base at least one is nondecreasing in its exponent, so

$$(\log x)^a \leq (\log x)^b, \qquad B_a(x) \leq B_b(x).$$

Multiplication by $\sqrt{x} \geq 0$ preserves the inequality. Thus each pair in $S_a$ with $x \geq N$ is also in $S_b$: the positivity and nonzero-gap conditions remain true and $D(x, y) < B_a(x) \leq B_b(x)$. By Lemma 2, that subset of $S_a$ is infinite if $I(a)$ holds. Therefore $S_b$ is infinite. □

**Theorem 2.** *The literal conjecture* (1) *is false.*

*Proof.* Assume $C$. Its first clause is $I(1/4)$. Theorem 1, with $a = 1/4$ and $b = 1$, gives $I(1)$. Since $1 > 1/4$, the second clause of $C$ gives $\neg I(1)$, a contradiction. □

This is a negation of a conjunction, not a construction of an infinite approximation family and not a proof that $S_{1/4}$ is finite. In fact the stronger implication $I(1/4) \Rightarrow I(b)$ is established for every $b \geq 1/4$. The choice of natural logarithm only fixes the finite cutoff; the same argument works with any fixed logarithm base greater than one.

## 4 Lean correspondence and reproduction

The project uses Lean 4.19.0 and Mathlib v4.19.0, with all transitive package revisions pinned in `lean/lake-manifest.json`. The mathematical source is `lean/HallReverse.lean`. In namespace `HallReverse`, `gap`, `upperBound`, `admissible`, and `infinitelyMany` define $D$, $B$, $S$, and $I$ directly using `Real.sqrt`, `Real.log`, real powers, and `Set.Infinite`. The definitions `cannotIncrease` and `conjecture` give the second clause and the full conjunction in (1).

The lemmas `finite_fiber`, `finite_below`, and `infinite_tail` prove the finite-exception argument. The theorem `infinitelyMany_mono` is Theorem 1. The final theorem is

`HallReverse.conjecture_false : ¬ HallReverse.conjecture`.

It has no extra assumptions. The source prints axiom dependencies for all seven principal theorems.

`lean/Verification.lean` contains 13 kernel-checked examples: both signs of the integer gap; exclusion of zero coordinates; exclusion of every perfect cube–square coincidence; the zero bound and exclusion at $x = 1$; the fully expanded admissibility predicate; infinitude beyond any cutoff; two specific larger exponents; general exponent monotonicity; and the negation of the conjunction with every defining expression exposed. Both modules are default Lake targets. From `lean/`, a standard reproduction is:

```
lake exe cache get
lake --wfail build
lake env lean -DwarningAsError=true HallReverse.lean
lake env lean -DwarningAsError=true Verification.lean
```

The package includes local verification records and an independent AI-agent semantic review. These are contributor checks, not maintainer acceptance. Verification covers a fresh build of the submitted modules, direct source replay with warnings treated as errors, dependency-closure and axiom checks, and explicit positive and negative controls for the auditing tool. The only permitted foundational axioms are `propext`, `Classical.choice`, and `Quot.sound`; there are no admitted proofs, custom axioms, or `native_decide` proofs in the submitted mathematical modules. Archived negative controls intentionally contain rejected constructs to test the auditing tools; they are outside the mathematical proof library. No auxiliary numerical computation is needed for the mathematics.

Pinned dependency sources and supplied cached imports are checked by identity; the complete Mathlib library is not rebuilt, and the supplied Lean executable is not bootstrapped. The source, tests, report, and recorded file hashes allow the local validation to be distinguished from those trust assumptions.